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United States Patent	12409590
Kind Code	B2
Date of Patent	September 09, 2025
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Manufacturing apparatus, manufacturing system, and manufacturing method

Abstract

A manufacturing apparatus includes a defining portion, a plunger, and a valve member. The defining portion defines a supply port, a discharge port, and a resin channel. The resin channel includes a reserving portion, a first partial channel connecting the supply port and the reserving portion, a second partial channel connecting the reserving portion and the discharge port, a relay channel connecting the first partial channel, the reserving portion and the second partial channel, and a connecting channel connecting the first partial channel and the second partial channel. The plunger discharges the molten resin reserved in the reserving portion. The valve member is provided in the connecting channel and the second partial channel. The first partial channel and the relay channel are connected to the connecting channel so as to intersect with the connecting channel. The valve member includes a first valve portion and a second valve portion.

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Appl. No.: **18/179524**

Filed: **March 07, 2023**

Prior Publication Data

Document Identifier	Publication Date
US 20230302697 A1	Sep. 28, 2023

Foreign Application Priority Data

JP	2022-046727	Mar. 23, 2022

Publication Classification

Int. Cl.: **B29C45/23** (20060101); **B29C45/18** (20060101); **B29C45/26** (20060101); **B29C45/53** (20060101)

U.S. Cl.:

CPC **B29C45/23** (20130101); **B29C45/18** (20130101); **B29C45/535** (20130101); B29C2045/2683 (20130101)

Field of Classification Search

CPC: B29C (45/23); B29C (45/18); B29C (45/535); B29C (45/2073); B29C (45/27); B29C (45/28); B29C (45/231); B29C (45/2806); B29C (45/53); B29C (2045/2683)

USPC: 425/561

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Background/Summary

BACKGROUND OF THE INVENTION
Field of the Invention
(1) The present disclosure relates to manufacturing of a product.
Description of the Related Art
(2) An injection molding apparatus capable of manufacturing a product by molding a plurality of molded articles or molding a plurality of molded articles on a workpiece by a single injection operation is generally known. To mold a plurality of molded articles simultaneously, a plurality of cavities corresponding to the shapes of the molded articles are defined in a mold, and molten resin is injected into the plurality of cavities simultaneously. If the plurality of cavities have the same shape, a plurality of molded articles having the same shape can be manufactured, and if the plurality of cavities have different shapes, a plurality of molded articles having different shapes can be manufactured. In addition, a molded article is manufactured by injecting molten resin into a single cavity through a plurality of discharge ports. In any of these cases, molten resin is discharged through a plurality of discharge ports. In the case of discharging molten resin through a plurality of discharge ports, the amount of molten resin discharged through each discharge port needs to be controlled.
(3) Japanese Patent Laid-Open No. 2001-205656 discloses an injection molding apparatus in which discharge ports connected to respective cavities are each provided with a valve pin. In the injection molding apparatus described in Japanese Patent Laid-Open No. 2001-205656, the opening/closing timing of the discharge port by the valve pin is controlled in accordance with the capacity of each cavity, and thus the amount of molten resin injected into each cavity is controlled.
(4) Each valve pin is configured to be driven by a driving mechanism including an electric motor, an air cylinder, or the like. In the case of controlling the amount or pressure of resin supplied to each cavity by controlling the opening/closing timing of each discharge port as described in Japanese Patent Laid-Open No. 2001-205656, an error such as a time lag between opening/closing timings of the discharge ports can occur. This error can cause variations in the amount of resin injected into respective cavities, or cause difference in the resin pressure in each cavity, which can affect the quality of the molded article. Similarly, also in the case of injecting molten resin into a single cavity through a plurality of discharge ports, the quality of the molded article can be affected.
SUMMARY OF THE INVENTION
(5) According to a first aspect of the present invention, a manufacturing apparatus includes a defining portion, a plunger, and a valve member. The defining portion is configured to define a supply port through which molten resin is supplied, a discharge port through which the molten resin is discharged, and a resin channel connecting the supply port and the discharge port. The resin channel includes a reserving portion configured to reserve the molten resin, a first partial channel connecting the supply port and the reserving portion, a second partial channel connecting the

reserving portion and the discharge port, a relay channel connecting the first partial channel and the reserving portion and connecting the reserving portion and the second partial channel, and a connecting channel connecting the first partial channel and the second partial channel. The plunger is movable to change a capacity of the reserving portion and is configured to discharge the molten resin reserved in the reserving portion from the discharge port by moving to reduce the capacity of the reserving portion. The valve member is provided in the connecting channel and the second partial channel and is movable in a channel direction of the second partial channel. The first partial channel and the relay channel are connected to the connecting channel so as to intersect with the connecting channel. The valve member includes a first valve portion configured to open and close the discharge port by moving in the channel direction, and a second valve portion configured to, in a case where the first valve portion has moved to a position to close the discharge port, open an exit of the molten resin of the first partial channel such that the first partial channel communicates with the relay channel, and in a case where the first valve portion has moved to a position to open the discharge port, close the exit of the first partial channel.

(6) According to a second aspect of the present invention, a manufacturing apparatus includes a defining portion, a first plunger, a second plunger a first valve member, and a second valve member. The defining portion is configured to define a supply port through which molten resin is supplied, a first discharge port through which the molten resin is discharged, a second discharge port through which the molten resin is discharged, a first resin channel connecting the supply port and the first discharge port, and a second resin channel connecting the supply port and the second discharge port. The first resin channel includes a first reserving portion configured to reserve the molten resin. The second resin channel includes a second reserving portion configured to reserve the molten resin. The first plunger is movable to change a capacity of the first reserving portion and is configured to discharge the molten resin reserved in the first reserving portion from the first discharge port by moving to reduce the capacity of the first reserving portion. The second plunger is movable to change a capacity of the second reserving portion and is configured to discharge the molten resin reserved in the second reserving portion from the second discharge port by moving to reduce the capacity of the second reserving portion. The first resin channel includes a first partial channel connecting the supply port and the first reserving portion, a second partial channel connecting the first reserving portion and the first discharge port, and a first relay channel connecting the first partial channel and the first reserving portion and connecting the first reserving portion and the second partial channel. The second resin channel includes a third partial channel connecting the supply port and the second reserving portion, a fourth partial channel connecting the second reserving portion and the second discharge port, and a second relay channel connecting the third partial channel and the second reserving portion and connecting the second reserving portion and the fourth partial channel. The first plunger discharges the molten resin reserved in the first reserving portion from the first discharge port via the first relay channel and the second partial channel by moving to reduce a capacity of the first reserving portion. The second plunger discharges the molten resin reserved in the second reserving portion from the second discharge port via the second relay channel and the fourth partial channel by moving to reduce a capacity of the second reserving portion. The first valve member includes a first valve portion configured to open and close the first discharge port in a case where the first valve member moves, and a second valve portion configured to, in a case where the first valve portion has moved to a position to close the first discharge port, open an exit of the molten resin of the first partial channel such that the first partial channel communicates with the first relay channel, and in a case where the first valve portion has moved to a position to open the first discharge port, close the exit of the first partial channel. The second valve member includes a third valve portion configured to open and close the second discharge port in a case where the second valve member moves, and a fourth valve portion configured to, in a case where the third valve portion has moved to a position to close the second discharge port, open an exit of the molten resin of the third partial channel such that the third

partial channel communicates with the second relay channel, and in a case where the third valve portion has moved to a position to open the second discharge port, close the exit of the third partial channel. The first partial channel is longer than the third partial channel.
(7) According to a third aspect of the present invention, a manufacturing apparatus includes a defining portion, a first plunger, a second plunger, a first heater, and a second heater. The defining portion is configured to define a supply port through which molten resin is supplied, a first discharge port through which the molten resin is discharged, a second discharge port through which the molten resin is discharged, a first resin channel connecting the supply port and the first discharge port, and a second resin channel connecting the supply port and the second discharge port. The first resin channel includes a first reserving portion configured to reserve the molten resin, a first partial channel connecting the supply port and the first reserving portion, and a second partial channel connecting the first reserving portion and the first discharge port. The second resin channel includes a second reserving portion configured to reserve the molten resin, a third partial channel connecting the supply port and the second reserving portion, and a fourth partial channel connecting the second reserving portion and the second discharge port. The first plunger is movable to change a capacity of the first reserving portion and is configured to discharge the molten resin reserved in the first reserving portion from the first discharge port by moving to reduce the capacity of the first reserving portion. The second plunger is movable to change a capacity of the second reserving portion and is configured to discharge the molten resin reserved in the second reserving portion from the second discharge port by moving to reduce the capacity of the second reserving portion. The first heater is disposed along the second partial channel. The second heater is disposed along the fourth partial channel. The first plunger discharges the molten resin reserved in the first reserving portion from the first discharge port via the second partial channel by moving to reduce a capacity of the first reserving portion. The second plunger discharges the molten resin reserved in the second reserving portion from the second discharge port via the fourth partial channel by moving to reduce a capacity of the second reserving portion. The second partial channel is longer than the fourth partial channel. The first heater is longer than the second heater.
(8) Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a perspective view of a manufacturing apparatus according to a first embodiment.
(2) FIG. **2** is a longitudinal section view of an injection unit according to the first embodiment.
(3) FIG. **3**A is a plan view of the injection unit according to the first embodiment.
(4) FIG. **3**B is a plan view of the injection unit according to the first embodiment.
(5) FIG. **4** is an explanatory diagram of two resin channels according to the first embodiment.
(6) FIG. **5** is a perspective view of part of a valve pin according to the first embodiment.
(7) FIG. **6**A is a section view of a second valve portion and the vicinity thereof according to the first embodiment.
(8) FIG. **6**B is a section view of the second valve portion and the vicinity thereof according to the first embodiment.
(9) FIG. **7**A is a section view of the second valve portion and the vicinity thereof according to the first embodiment.
(10) FIG. **7**B is a section view of the second valve portion and the vicinity thereof according to the first embodiment.
(11) FIG. **8** is a longitudinal section view of the injection unit according to the first embodiment.
(12) FIG. **9** is a longitudinal section view of the injection unit according to the first embodiment.

(13) FIG. **10** is a longitudinal section view of the injection unit according to the first embodiment.
(14) FIG. **11** is a longitudinal section view of the injection unit according to the first embodiment.
(15) FIG. **12** is a longitudinal section view of the injection unit according to the first embodiment.
(16) FIG. **13** is an explanatory diagram of a product according to the first embodiment.
(17) FIG. **14** is an explanatory diagram of the product according to the first embodiment.
(18) FIG. **15** is a perspective view of a manufacturing system according to a second embodiment.
(19) FIG. **16** is an explanatory diagram of a product according to the second embodiment.
(20) FIG. **17** is a plan view of part of a manufacturing apparatus according to a third embodiment.
(21) FIG. **18** is a plan view of part of a manufacturing apparatus according to a fourth embodiment.
(22) FIG. **19** is a plan view of part of a manufacturing apparatus according to a fifth embodiment.
(23) FIG. **20** is an explanatory diagram of two resin channels of a manufacturing apparatus according to a sixth embodiment.

DESCRIPTION OF THE EMBODIMENTS

(24) Exemplary embodiments of the present disclosure will be described below in detail with reference to drawings. The manufacturing apparatus according to the present disclosure can be applied to an injection molding apparatus that molds a product by injecting molten resin into a mold, and an injection molding apparatus in which a workpiece such as a molded part or a metal part is set in the apparatus and a product is molded by injecting molten resin onto the workpiece. For example, the product can be manufactured by insert molding, outsert molding, or two-color molding. Hereinafter, outsert molding will be described as an example.

First Embodiment

(25) FIG. **1** is a perspective view of a manufacturing apparatus **1000** according to a first embodiment. The manufacturing apparatus **1000** is, for example, a small injection molding apparatus, and can be used in a manufacturing line for a product. To mold a small molded article, an injection molding apparatus of a pre-plunger type is suitable as a small injection molding apparatus. A case where the manufacturing apparatus **1000** is an injection molding apparatus of a pre-plunger type will be described. In an injection molding apparatus of a pre-plunger type, the amount of molten resin to be injected into a cavity is measured by reserving the molten resin in a cylinder defining a reserving portion.

(26) The manufacturing apparatus **1000** of the first embodiment is used for manufacturing a product by outsert molding of a molded article at each of a plurality of positions on a workpiece **13**. The product manufactured by the manufacturing apparatus **1000** may be an intermediate product or a final product.

(27) The manufacturing apparatus **1000** includes an injection unit **100**, a holding member **30** that holds the workpiece **13**, and a control apparatus **200** that controls the injection unit **100**. The control apparatus **200** is an example of a controller, and is constituted by, for example, one or a plurality of computers.

(28) FIG. **2** is a longitudinal section view of the injection unit **100** according to the first embodiment. The injection unit **100** includes a resin supply portion **46**, a plasticizing portion **49**, a defining portion **52**, and an injection portion **40**. The resin supply portion **46** is, for example, a hopper. For example, a resin material of a pellet shape is charged into the resin supply portion **46**. For example, the resin material is a thermoplastic resin. The resin material charged into the resin supply portion **46** is supplied to the plasticizing portion **49**.

(29) The plasticizing portion **49** is connected to a supply port S**1** through which molten resin is supplied. The plasticizing portion **49** includes a cylinder **47** having a hollow cylindrical shape, a screw **50** disposed inside the cylinder **47**, and a heater **51** disposed on the outer peripheral surface of the cylinder **47**. A resin material is supplied into the cylinder **47** from the resin supply portion **46**. The cylinder **47** is heated by the heater **51**, and thus the resin material supplied into the cylinder **47** is heated. The resin material is heated, and thus conveyed by the screw **50** to the defining portion **52** while being plasticized and melted. The defining portion **52** includes a plurality of

members defining a plurality of resin channels. The resin melted by the plasticizing portion **49**, that is, molten resin is supplied to each resin channel. The direction in which the resin flows in the channel will be referred to as a channel direction. Typically, the channel direction of a channel coincides with the longitudinal direction of the channel. If the channel has a cylindrical shape, the channel direction of the channel coincides with the axial direction of the cylindrical shape.
(30) FIGS. **3**A and **3**B are plan views of the injection unit **100** according to the first embodiment. In each of FIGS. **3**A and **3**B, part of the injection unit **100** is illustrated in section view. In the first embodiment, as an example of a plurality of resin channels, three resin channels **521**, **522**, and **523** through which molten resin flows from a supply port S**1** connected to the single plasticizing portion **49** to three discharge ports G**1**, G**2**, and G**3** are defined by the defining portion **52**. To be noted, the number of the resin channels is not limited to 3, and may be 2, 4, or more.
(31) The resin channel **521** is an example of a first resin channel, and is a resin channel connecting the discharge port G**1** and the supply port S**1** connected to the plasticizing portion **49**. The resin channel **522** is an example of a second resin channel, and is a resin channel connecting the discharge port G**2** and the supply port S**1** connected to the plasticizing portion **49**. The resin channel **523** is an example of a third resin channel, and is a resin channel connecting the discharge port G**3** and the supply port S**1** connected to the plasticizing portion **49**. In the first embodiment, a main channel **52***a* shared by the plurality of resin channels **521**, **522**, and **523** is defined by the defining portion **52**. The main channel **52***a* is connected to the supply port S**1** connected to the plasticizing portion **49**.
(32) The discharge ports G**1**, G**2**, and G**3** are each a gate through which the molten resin is discharged. The discharge port G**1** is an example of a first discharge port. The discharge port G**2** is an example of a second discharge port. The discharge port G**3** is an example of a third discharge port. The discharge port G**1** is connected to a cavity CV**1**. The discharge port G**2** is connected to a cavity CV**2**. The discharge port G**3** is connected to a cavity CV**3**. The cavity CV**1** is an example of a first cavity. The cavity CV**2** is an example of a second cavity. The cavity CV**3** is an example of a third cavity. The molten resin discharged from the discharge port G**1** is injected into the cavity CV**1**, the molten resin discharged from the discharge port G**2** is injected into the cavity CV**2**, and the molten resin discharged from the discharge port G**3** is injected into the cavity CV**3**. The defining portion **52** is overall heated by the heater. For example, the cavities CV**1**, CV**2**, and CV**3** are each defined by a mold and the workpiece **13**.
(33) FIG. **4** is an explanatory diagram of the two resin channels **521** and **522** according to the first embodiment. To be noted, in FIG. **4**, illustration of the resin channel **523** is omitted.
(34) The resin channel **521** includes a reserving portion **58**, a partial channel **52***x*, a partial channel **52***e*, and a partial channel **52***h*. The reserving portion **58** is an example of a first reserving portion, and is a space capable of reserving the molten resin supplied from the supply port S**1** connected to the plasticizing portion **49**. The partial channel **52***x* is an example of a first partial channel, and connects the reserving portion **58** and the supply port S**1** connected to the plasticizing portion **49**. The partial channel **52***x* includes the main channel **52***a* and a branch channel **52***b*. The branch channel **52***b* is an example of a first branch channel, and branches from the main channel **52***a*. The partial channel **52***e* is an example of a first relay channel, connects the reserving portion **58** and the branch channel **52***b* of the partial channel **52***x*, and connects the reserving portion **58** and the partial channel **52***h*. The partial channel **52***h* is an example of a second partial channel, and connects the partial channel **52***e* and the discharge port G**1**. That is, the partial channel **52***h* connects the reserving portion **58** and the discharge port G**1**.
(35) The resin channel **522** includes a reserving portion **59**, a partial channel **52***y*, a partial channel **52***f*, and a partial channel **52***i*. The reserving portion **59** is an example of a second reserving portion, and is a space capable of reserving the molten resin supplied from the supply port S**1** connected to the plasticizing portion **49**. The partial channel **52***y* is an example of a third partial channel, and connects the reserving portion **59** and the supply port S**1** connected to the plasticizing portion **49**.

The partial channel **52***y* includes the main channel **52***a* and a branch channel **52***c*. The branch channel **52***c* is an example of a second branch channel, and branches from the main channel **52***a*. The partial channel **52***f* is an example of a second relay channel, connects the reserving portion **59** and the branch channel **52***c* of the partial channel **52***y*, and connects the reserving portion **59** and the partial channel **52***i*. The partial channel **52***i* is an example of a fourth partial channel, and connects the partial channel **52***f* and the discharge port G**2**. That is, the partial channel **52***i* connects the reserving portion **59** and the discharge port G**2**.

(36) The resin channel **523** includes a reserving portion **60**, a partial channel including the main channel **52***a* and a branch channel **52***d*, a partial channel **52***g*, and a partial channel **52***j*. The reserving portion **60** is an example of a third reserving portion, and is a space capable of reserving the molten resin supplied from the supply port S**1** connected to the plasticizing portion **49**. The partial channel including the main channel **52***a* and the branch channel **52***d* is an example of a fifth partial channel, and connects the reserving portion **60** and the supply port S**1** connected to the plasticizing portion **49**. The branch channel **52***d* is an example of a third branch channel, and branches from the main channel **52***a*. The partial channel **52***g* is an example of a third relay channel, connects the reserving portion **60** and the branch channel **52***d*, and connects the reserving portion **60** and the partial channel **52***j*. The partial channel **52***j* is an example of a sixth partial channel, and connects the partial channel **52***g* and the discharge port G**3**. That is, the partial channel **52***j* connects the reserving portion **60** and the discharge port G**3**.

(37) As described above, the main channel **52***a* shared by the three resin channels **521**, **522**, and **523** is defined inside the defining portion **52**. In addition, the three branch channels **52***b*, **52***c*, and **52***d* branching from the main channel **52***a* are defined inside the defining portion **52**. The branch channels **52***b*, **52***c*, and **52***d* are each defined in a bent shape so as not to be linearly connected to the main channel **52***a*. As a result, a situation in which the flow of molten resin is concentrated on part of the branch channels, that is, a situation in which the resin pressure in part of the branch channels is too high can be suppressed.

(38) In addition, the branch channels **52***b*, **52***c*, and **52***d* are each designed to have a different length. That is, as illustrated in FIG. **4**, the total length L**1** of the partial channel **52***x* and the partial channel **52***e* is different from the total length L**2** of the partial channel **52***y* and the partial channel **52***f*.

(39) The length of the partial channel **52***e* is larger than the diameter of the partial channel **52***e*. Further, the partial channel **52***e* is defined such that the sectional area of the channel increases stepwise toward the reserving portion **58**. In addition, the length of the partial channel **52***f* is larger than the diameter of the partial channel **52***f*. Further, the partial channel **52***f* is defined such that the sectional area of the channel increases stepwise toward the reserving portion **59**. The length of the partial channel **52***g* is larger than the diameter of the partial channel **52***g*. Further, the partial channel **52***g* is defined such that the sectional area of the channel increases stepwise toward the reserving portion **60**. As a result of this, the reserving portions **58**, **59**, and **60** can smoothly reserve the molten resin.

(40) To be noted, the partial channel **52***e* may be defined such that the sectional area of the channel continuously increases toward the reserving portion **58**. In addition, the partial channel **52***f* may be defined such that the sectional area of the channel continuously increases toward the reserving portion **59**. In addition, the partial channel **52***g* may be defined such that the sectional area of the channel continuously increases toward the reserving portion **60**.

(41) The injection portion **40** includes plungers **55**, **56**, and **57**, a driving portion **70**, valve pins **61**, **62**, and **63**, and an unillustrated driving unit that drives the valve pins **61**, **62**, and **63**.

(42) The plunger **55** is an example of a first plunger, and is disposed in a cylinder of the defining portion **52** so as to be movable to change the capacity of the reserving portion **58**. The plunger **55** is capable of injecting the molten resin reserved in the reserving portion **58** into the cavity CV**1** from the discharge port G**1** via the partial channels **52***e* and **52***h* by moving to reduce the capacity of the

reserving portion **58**.
(43) The plunger **56** is an example of a second plunger, and is disposed in a cylinder of the defining portion **52** so as to be movable to change the capacity of the reserving portion **59**. The plunger **56** is capable of injecting the molten resin reserved in the reserving portion **59** into the cavity CV**2** from the discharge port G**2** via the partial channels **52***f* and **52***i* by moving to reduce the capacity of the reserving portion **59**.
(44) The plunger **57** is an example of a third plunger, and is disposed in a cylinder of the defining portion **52** so as to be movable to change the capacity of the reserving portion **60**. The plunger **57** is capable of injecting the molten resin reserved in the reserving portion **60** into the cavity CV**3** from the discharge port G**3** via the partial channels **52***g* and **52***j* by moving to reduce the capacity of the reserving portion **60**.
(45) As described above, the molten resin can be supplied to the plurality of cavities CV**1**, CV**2**, and CV**3** via the plurality of resin channels **521**, **522**, and **523** from the single plasticizing portion **49**, that is, from the single supply port S**1**. Further, the resin channel **521** includes the reserving portion **58**, the resin channel **522** includes the reserving portion **59**, the resin channel **523** includes the reserving portion **60**, the reserving portion **58** includes the plunger **55**, the reserving portion **59** includes the plunger **56**, and the reserving portion **60** includes the plunger **57**. As a result of this, molten resin corresponding to respective capacities of the cavities CV**1**, CV**2**, and CV**3** can be respectively reserved in the reserving portions **58**, **59**, and **60**, and can be respectively injected into the cavities CV**1**, CV**2**, and CV**3** by the plungers **55**, **56**, and **57**. As a result of this, molded articles respectively molded by the cavities CV**1**, CV**2**, and CV**3** can be manufactured with high precision, and thus the quality of each molded article is improved. In the first embodiment, the plurality of molded articles molded on the workpiece **13** can be manufactured with high precision, and thus the quality of the manufactured product is improved. In addition, in the first embodiment, the resin channels **521**, **522**, and **523** are configured such that the molten resin is supplied from the single plasticizing portion **49** to the reserving portions **58**, **59**, and **60**. As a result of this, a plurality of plasticizing portions corresponding to the plurality of reserving portions **58**, **59**, and **60** do not need to be provided, and thus the size of the manufacturing apparatus **1000**, that is, the size of the injection unit **100** can be reduced.
(46) The driving portion **70** is capable of individually driving the plungers **55**, **56**, and **57**. In the first embodiment, the driving portion **70** includes a driving mechanism **71** capable of driving the plunger **55**, a driving mechanism **72** capable of driving the plunger **56**, and a driving mechanism **73** capable of driving the plunger **57**. The driving mechanisms **71**, **72**, and **73** are capable of operating independently from each other. The driving portion **70**, that is, the driving mechanisms **71**, **72**, and **73** operate by being controlled by the control apparatus **200**. The driving mechanisms **71**, **72**, and **73** each include, for example, an electric mechanism such as a motor, or an air pressure mechanism such as an air cylinder.
(47) The driving mechanism **71** includes a pressing member **711** capable of coming into contact with and out of contact from the plunger **55**. The pressing member **711** is capable of pressing the plunger **55** so as to reduce the capacity of the reserving portion **58**. The driving mechanism **72** includes a pressing member **712** capable of coming into contact with and out of contact from the plunger **56**. The pressing member **712** is capable of pressing the plunger **56** so as to reduce the capacity of the reserving portion **59**. The driving mechanism **73** includes a pressing member **713** capable of coming into contact with and out of contact from the plunger **57**. The pressing member **713** is capable of pressing the plunger **57** so as to reduce the capacity of the reserving portion **60**. The pressing member **711** is an example of a first pressing member. The pressing member **712** is an example of a second pressing member. The pressing member **713** is an example of a third pressing member.
(48) The valve pins **61**, **62**, and **63** are valve members whose distal ends are valves (gate valves) that respectively open and close the discharge ports G**1**, G**2**, and G**3**. The valve pin **61** is an

example of a first valve member. The valve pin **62** is an example of a second valve member. The valve pin **63** is an example of a third valve member. As illustrated in FIG. **1**, the valve pin **61** is a rod-like member extending in an axial direction Z**1**. The valve pin **62** is a rod-like member extending in an axial direction Z**2**. The valve pin **63** is a rod-like member extending in an axial direction Z**3**. Although the axial directions Z**1**, Z**2**, and Z**3** are parallel to each other in the first embodiment, these directions may intersect with each other.
(49) The distal end of the valve pin **61** in the axial direction Z**1** is a valve portion **61***a* that opens and closes the discharge port G**1** by moving in the axial direction Z**1**. The distal end of the valve pin **62** in the axial direction Z**2** is a valve portion **62***a* that opens and closes the discharge port G**2** by moving in the axial direction Z**2**. The distal end of the valve pin **63** in the axial direction Z**3** is a valve portion that opens and closes the discharge port G**3** by moving in the axial direction Z**3**. The valve pins **61**, **62**, and **63** are respectively driven in the axial directions Z**1**, Z**2**, and Z**3** by an unillustrated driving unit. The unillustrated driving unit includes, for example, an electric mechanism such as a motor, or an air pressure mechanism such as an air cylinder.
(50) The valve pin **61** is provided in the partial channel **52***h*, the valve pin **62** is provided in the partial channel **52***i*, and the valve pin **63** is provided in the partial channel **52***j*. The distal end of the partial channel **52***h* in the axial direction Z**1** is the discharge port G**1**. The distal end of the partial channel **52***i* in the axial direction Z**2** is the discharge port G**2**. The distal end of the partial channel **52***j* in the axial direction Z**3** is the discharge port G**3**.
(51) The partial channel **52***h* is a channel used for supplying the molten resin from the reserving portion **58** to the cavity CV**1**. The partial channel **52***i* is a channel used for supplying the molten resin from the reserving portion **59** to the cavity CV**2**. The partial channel **52***j* is a channel used for supplying the molten resin from the reserving portion **60** to the cavity CV**3**. The injection unit **100** includes a heater **41** illustrated in FIG. **1** disposed along the partial channel **52***h*, a heater **42** illustrated in FIG. **1** disposed along the partial channel **52***i*, and a heater **43** illustrated in FIG. **1** disposed along the partial channel **52***j*. The heater **41** is an example of a first heater. The heater **42** is an example of a second heater. The heater **43** is an example of a third heater. The heater **41** is disposed on the outer peripheral surface of a protruding member **91** having a cylindrical shape and defining the partial channel **52***h* in the defining portion **52**. The heater **42** is disposed on the outer peripheral surface of a protruding member **92** having a cylindrical shape and defining the partial channel **52***i* in the defining portion **52**. The heater **43** is disposed on the outer peripheral surface of a protruding member **93** having a cylindrical shape and defining the partial channel **52***j* in the defining portion **52**.
(52) The positions of the discharge ports G**1**, G**2**, and G**3** in a height direction, that is, the lengths of the partial channels **52***h*, **52***i*, and **52***j* are set in accordance with the positions of the cavities CV**1**, CV**2**, and CV**3**, that is, the positions where the molded articles are to be molded on the workpiece **13**. Therefore, the partial channels **52***h*, **52***i*, and **52***j* can be set to different lengths. Therefore, the lengths of the heaters **41**, **42**, and **43** are respectively set in accordance with the lengths of the partial channels **52***h*, **52***i*, and **52***j*.
(53) In the example of the first embodiment, the partial channel **52***h* is longer than the partial channel **52***i*. Therefore, the heater **41** is set to be longer than the heater **42**. In the first embodiment, the length of the partial channel **52***h* is the length of the partial channel **52***h* in the axial direction Z**1**, and the length of the partial channel **52***i* is the length of the partial channel **52***i* in the axial direction Z**2**. In addition, in the first embodiment, the length of the heater **41** is the length of the heater **41** in the axial direction Z**1**, and the length of the heater **42** is the length of the heater **42** in the axial direction Z**2**. When the length of the partial channel **52***h* is L**11** and the length of the partial channel **52***i* is L**12**, L**11**>L**12** holds. In addition, when the length of the heater **41** is L**21** and the length of the heater **42** is L**22**, L**21**>L**22** holds.
(54) The molten resin that flows in the partial channels **52***h*, **52***i*, and **52***j* and is supplied to the cavities CV**1**, CV**2**, and CV**3** is held at a predetermined temperature by the heaters **41**, **42**, and **43**,

and thus a good molten state can be maintained.
(55) The valve pin **61** includes a valve portion **61***b* that opens and closes an exit of the molten resin of the partial channel **52***x*, that is, opens and closes an exit **81** of the molten resin of the branch channel **52***b*. The valve pin **62** includes a valve portion **62***b* that opens and closes an exit of the molten resin of the partial channel **52***y*, that is, opens and closes an exit **82** of the molten resin of the branch channel **52***c*. The valve section **62***b* has substantially the same configuration as the valve section **61***b*. To be noted, although the illustration thereof is omitted, the valve pin **63** includes a valve portion having a similar configuration to the valve portions **61***b* and **62***b* that opens and closes an exit **83** of the molten resin of the branch channel **52***d*.
(56) The configurations of the valve portion **62***b* of the valve pin **62** and the valve portion of the valve pin **63** are similar to the configuration of the valve portion **61***b* of the valve pin **61**, and therefore the valve pin **61***b* will be described below. Here, the valve portion **61***a* of the valve pin **61** is an example of a first valve portion, and the valve portion **61***b* of the valve pin **61** is an example of a second valve portion. The valve portion **62***a* of the valve pin **62** is an example of a third valve portion, and the valve portion **62***b* of the valve pin **62** is an example of a fourth valve portion.
(57) FIG. **5** is a perspective view of part of the valve pin **61** according to the first embodiment. FIG. **6**A is a section view of the valve portion **61***b* and the vicinity thereof in FIG. **2**. FIG. **6**A illustrates a state in which the valve portion **61***a* of the valve pin **61** has moved to a closed position P**1** to close the discharge port G**1**. FIG. **6**B is a section view of the valve portion **61***b* and the vicinity thereof taken along a line VIB-VIB of FIG. **6**A. FIG. **7**A is a section view of the valve portion **61***b* and the vicinity thereof in FIG. **2**. FIG. **7**A illustrates a state in which the valve portion **61***a* of the valve pin **61** has moved to an open position P**2** to close the discharge port G**1**. FIG. **7**B is a section view of the valve portion **61***b* and the vicinity thereof taken along a line VIIB-VIIB of FIG. **7**A.
(58) The defining portion **52** includes an inner peripheral surface **155** defining a hole H**1** extending in the axial direction Z**1**. The inner peripheral surface **155** is also a surface defining a connecting channel **52***k* and the partial channel **52***h*. The connecting channel **52***k* is a channel connecting the partial channel **52***x* and the partial channel **52***h*. The valve pin **61** is inserted in the hole H**1** defined by the inner peripheral surface **155**, and is movable in the axial direction Z**1** along the inner peripheral surface **155**. Therefore, the valve portion **61***b* that is part of the valve pin **61** is slidable in the axial direction Z**1** in the hole H**1** defined by the inner peripheral surface **155**.
(59) Part of the hole H**1** is the partial channel **52***h* and the connecting channel **52***k*. In addition, the branch channel **52***b* of the partial channel **52***x* and the partial channel **52***e* are connected to the hole H**1**, that is, connected to the connecting channel **52***k* while intersecting with the connecting channel **52***k*. In the first embodiment, the branch channel **52***b* and the partial channel **52***e* are orthogonally connected to the hole H**1**, that is, to the connecting channel **52***k*. In the inner peripheral surface **155**, the exit of the molten resin of the partial channel **52***x*, that is, the resin exit **81** of the molten resin of the branch channel **52***b* is provided.
(60) The valve pin **61**, that is, the valve portion **61***a* and the valve portion **61***b* are slidable in, among the axial direction Z**1**, a direction Z**11** for moving away from the discharge port G**1** and a direction Z**12** for moving closer to the discharge port G**1**. The direction Z**12** is opposite to the direction Z**11**. As a result of the valve pin **61** sliding in the axial direction Z**1**, the valve portion **61***a* is capable of moving to the closed position P**1** where to close the discharge port G**1**, and an open position P**2** to open the discharge port G**1**. The valve portion **61***a* moves from the closed position P**1** to the open position P**2** by sliding in the direction Z**11**. In addition, the valve portion **61***a* moves from the open position P**2** to the closed position P**1** by sliding in the direction Z**12**.
(61) The valve pin **61** includes a columnar portion **161** extending in the axial direction Z**1**, and a columnar portion **162** extending in the axial direction Z**1**. The valve portion **61***b* is provided between the columnar portions **161** and **162**. The columnar portion **161** is disposed on the discharge port G**1** side with respect to the columnar portion **162** and the valve portion **61***b*, and the

distal end of the columnar portion **161** in the axial direction Z**1** opens and closes the discharge port G**1** when the valve pin **61** moves in the axial direction Z**1**. That is, the distal end of the columnar portion **161** in the axial direction Z**1** serves as the valve portion **61***a* that opens and closes the discharge port G**1**.

(62) The columnar portion **162** has a larger diameter than the columnar portion **161**. The columnar portion **162** is fitted to the inner peripheral surface **155** of the defining portion **52** so as to be slidable in the axial direction Z**1**. That is, an outer peripheral surface **1621** of the columnar portion **162** is in contact with the inner peripheral surface **155** of the defining portion **52**.

(63) The columnar portion **161** is disposed so as not to be in contact with the inner peripheral surface **155** of the defining portion **52**, and a space between the inner peripheral surface **155** of the defining portion **52** and an outer peripheral surface **1611** of the columnar portion **161** serves as the partial channel **52***h*.

(64) The valve portion **61***b* includes a columnar portion **163** having the same diameter as the columnar portion **162**, and a columnar portion **164** having a smaller diameter than the columnar portions **162** and **163**. The columnar portions **163** and **164** are disposed between the columnar portion **161** and the columnar portion **162**. An outer peripheral surface **1631** of the columnar portion **163** is a columnar surface, and is in contact with the inner peripheral surface **155** of the defining portion **52** so as to be slidable in the axial direction Z**1**.

(65) The columnar portion **164** is disposed between the columnar portion **163** and the columnar portion **162**. Therefore, an annular groove is defined between the columnar portions **162** and **163**, and an annular space R**1** corresponding to the annular groove is defined between the columnar portion **164** and the inner peripheral surface **155** of the defining portion **52**. A groove portion **1632** extending in the axial direction Z**1** is provided on the columnar portion **163** such that the space R**1** and the partial channel **52***h* communicate with each other. A space R**2** connected to the space R**1** and the partial channel **52***h* is defined between the groove portion **1632** and the inner peripheral surface **155** of the defining portion **52**. The connecting channel **52***k* includes the spaces R**1** and R**2**. To be noted, the columnar portions **161** to **164** of the valve pin **61** each may or may not have a hollow shape.

(66) When the valve portion **61***a* of the valve pin **61** moves to the closed position P**1** as illustrated in FIG. **6**A, the outer peripheral surface **1631** moves to a position to open the exit **81** of the branch channel **52***b*. As a result of this, the partial channel **52***x* communicates with the partial channel **52***e* of the partial channel **52***h* via the connecting channel **52***k*. At this time, since the discharge port G**1** is closed by the valve portion **61***a* of the valve pin **61**, molten resin M**1** having flowed through the branch channel **52***b* of the partial channel **52***x* flows into the reserving portion **58** via the connecting channel **52***k* and the partial channel **52***e*, and is thus reserved in the reserving portion **58**.

(67) In addition, when the valve portion **61***a* of the valve pin **61** moves to the open position P**2** as illustrated in FIG. **7**A, the space R**1**, that is, the connecting channel **52***k* is displaced in the direction Z**11** with respect to the exit **81**, and the outer peripheral surface **1631** moves to a position to close the exit **81** of the branch channel **52***b*. At this time, the discharge port G**1** is open, and therefore the molten resin M**1** reserved in the reserving portion **58** is discharged from the discharge port G**1** via the partial channel **52***e* and the partial channel **52***h* by moving the plunger **55** to reduce the capacity of the reserving portion **58**.

(68) In the first embodiment, the branch channel **52***b* opened and closed by the valve portion **61***b* is connected to the hole H**1**, that is, the connecting channel **52***k* so as to intersect therewith. Therefore, the outer peripheral surface **1631** of the valve portion **61***b* receives the pressure of the molten resin M**1** staying in the branch channel **52***b* not in the axial direction Z**1** but in a direction intersecting with the axial direction Z**1**. The axial direction Z**1** is a driving direction of the valve pin **61**. In the example of FIG. **7**A, the outer peripheral surface **1631** of the valve portion **61***b* receives the pressure of the molten resin M**1** staying in the branch channel **52***b* in a direction orthogonal to the axial direction Z**1**.

(69) Therefore, movement of the valve pin **61**, that is, the valve portion **61***b* in the axial direction Z**1**, specifically the direction Z**12** is suppressed by the pressure of the molten resin M**1** staying in the branch channel **52***b*. As a result of this, the valve portion **61***a* is held at the open position P**2** to open the discharge port G**1** with high precision, and in addition, leakage of the molten resin M**1** from the branch channel **52***b* to the partial channel **52***h* at the valve portion **61***b* can be suppressed.
(70) In addition, the outer peripheral surface **1631** is a columnar surface extending straight in the axial direction Z**1**, and thus can be easily manufactured with high precision. The inner peripheral surface **155** defining the hole H**1** is also a columnar surface extending straight in the axial direction Z**1**, and thus can be easily manufactured with high precision. Therefore, leakage of the molten resin between the branch channel **52***b* and the partial channels **52***e* and **52***h* through a gap between the outer peripheral surface **1631** and the inner peripheral surface **155** can be effectively reduced.
(71) As a result of the valve portion **61***b* having the configuration described above, the variation in the amount of molten resin injected from the discharge port G**1** into the cavity CV**1** with respect to the capacity of the cavity CV**1** is reduced, and the quality of the molded article to be molded, that is, the quality of the product is improved.
(72) In the manufacturing apparatus **1000** of the first embodiment, products are sequentially manufactured by repeating an amount measuring step, an injection step, and a cooling step. To be noted, in these steps, the operation of the plungers **56** and **57** is approximately the same as the operation of the plunger **55**, and the operation of the valve pins **62** and **63** is approximately the same as the operation of the valve pin **61**. Therefore, only the operation of the plunger **55** and the valve pin **61** will be described below, and description of the operation of the plungers **56** and **57** and the operation of the valve pins **62** and **63** will be omitted.
(73) FIG. **8** is a longitudinal section view of the injection unit **100** according to the first embodiment. FIG. **8** illustrates a state in which the amount measuring step is completed.
(74) Here, before the amount measuring step is started, the control apparatus **200** performs drive control of the valve pin **61** to move the valve portion **61***a* of the closed position P**1** and move the valve portion **61***b* to the open position. As a result of this, the valve portion **61***a* closes the discharge port G**1**, and the valve portion **61***b* opens the exit **81** of the branch channel **52***b*. The connecting channel **52***k* illustrated in FIG. **6**A connecting the branch channel **52***b* and the partial channel **52***e* is formed by the valve portion **61***b*. Further, in the amount measuring step, the molten resin M**1** discharged from the plasticizing portion **49** is injected into the reserving portion **58** through the main channel **52***a*, the branch channel **52***b*, the connecting channel **52***k*, and the partial channel **52***e*.
(75) FIG. **9** is a longitudinal section view of the injection unit **100** according to the first embodiment. FIG. **9** illustrates a state in which the injection step is started. After the injection of the molten resin M**1** into the reserving portion **58** is completed, that is, after the amount measuring step is completed, the control apparatus **200** performs drive control of the valve pin **61** to move the valve portion **61***a* to the open position P**2** and move the valve portion **61***b* to the closed position as illustrated in FIG. **9**. As a result of this, the valve portion **61***a* opens the discharge port G**1**, and the valve portion **61***b* closes the exit **81** of the branch channel **52***b*.
(76) FIG. **10** is a longitudinal section view of the injection unit **100** according to the first embodiment. FIG. **10** illustrates a state in which the injection step is completed. In the injection step, the control apparatus **200** controls the driving mechanism **71** to cause the plunger **55** to push out the molten resin M**1** injected into the reserving portion **58**. That is, the control apparatus **200** drives the plunger **55** to move the plunger **55** to reduce the capacity of the reserving portion **58**. By moving the plunger **55** in this manner, the molten resin M**1** reserved in the reserving portion **58** is injected into the cavity CV**1** from the discharge port G**1** via the partial channel **52***e* and the partial channel **52***h*.
(77) At this time, the valve portion **61***b* closes the branch channel **52***b*, and therefore the molten resin M**1** flowing back to the branch channel **52***b*, that is, to the plasticizing portion **49** can be

suppressed. Therefore, the molten resin M**1** injected into the reserving portion **58** and measured to a predetermined amount can be stably discharged from the discharge port G**1**.

(78) In addition, the partial channel **52***e* has such a shape that the sectional area thereof increases stepwise toward the plunger **55**, that is, such a shape that the sectional area thereof decreases stepwise toward the discharge port G**1**. As described above, as compared with a case where the channel suddenly becomes narrow, when driving the plunger **55**, the driving force thereof is efficiently transmitted to the molten resin M**1**, and the pressure loss of the molten resin M**1** in the section from the partial channel **52***e* to the discharge port G**1** is reduced. Therefore, the molten resin M**1** can be injected into the cavity CV**1** at a desired pressure.

(79) After the injection step is completed, the amount measuring step of measuring the amount of the molten resin for next injection is performed in parallel with a cooling step of cooling the molten resin M**1** injected into the cavity CV**1**.

(80) FIG. **11** is a longitudinal section view of the injection unit **100** according to the first embodiment. FIG. **11** illustrates a state in which the amount measuring step is started. As illustrated in FIG. **11**, the control apparatus **200** performs drive control of the valve pin **61** to move the valve portion **61***a* to the closed position P**1** and move the valve portion **61***b* to the open position. As a result of this, the valve portion **61***a* closes the discharge port G**1**, and the valve portion **61***b* opens the exit **81** of the branch channel **52***b*. The connecting channel **52***k* connecting the branch channel **52***b* and the partial channel **52***e* is formed by the valve portion **61***b*. In addition, the control apparatus **200** controls the driving mechanism **71** and retracts the pressing member **711** to a position away from the plunger **55**.

(81) Further, in the amount measuring step, the control apparatus **200** rotates the screw **50** of the plasticizing portion **49**. As a result of this, molten resin is discharged from the plasticizing portion **49**. The molten resin M**1** discharged from the plasticizing portion **49** is injected into the reserving portion **58** through the main channel **52***a*, the branch channel **52***b*, the connecting channel **52***k*, and the partial channel **52***e*. At this time, the plunger **55** is pressed by the molten resin M**1** supplied to the reserving portion **58**, and thus moves closer to the pressing member **711**.

(82) FIG. **12** is a longitudinal section view of the injection unit **100** according to the first embodiment. FIG. **12** illustrates a state in which the amount measuring step has been completed. As illustrated in FIG. **12**, the plunger **55** is pressed by the molten resin M**1**, and thus moves back to come into contact with the pressing member **711**. That is, when the molten resin M**1** in the reserving portion **58** reaches a predetermined volume, the plunger **55** comes into contact with the pressing member **711**. The predetermined volume is equal to the capacity of the cavity CV**1**. In the case where contact of the plunger **55** with the pressing member **711** is detected, the control apparatus **200** stops the rotation of the screw **50** of the plasticizing portion **49**, and completes the amount measurement.

(83) In the first embodiment, the plunger **55** is separated from the pressing member **711** of the driving mechanism **71**. Therefore, in the amount measuring step, the plunger **55** does not need to be forcibly retracted. That is, the plunger **55** retracts as a result of the pressing force of the molten resin M**1**. If the plunger is forcibly retracted, there is a possibility that the capacity of the reserving portion suddenly increases, thus sudden pressure change occurs in the molten resin in the reserving portion, and the quality of the molten resin in the reserving portion changes, air is mixed in the molten resin, or the like. There is a possibility that a molding failure such as a void or a silver streak occurs in a molded article formed from the molten resin affected by the sudden pressure change.

(84) In contrast, in the first embodiment, since the plunger **55** and the driving mechanism **71** are separated, the plunger **55** can be moved back by being pushed back by the molten resin M**1** after the pressing member **711** of the driving mechanism **71** is retracted. As a result of this, occurrence of sudden pressure change of the molten resin M**1** in the reserving portion **58** can be reduced. Therefore, change in the quality of the molten resin M**1** or mixing of air in the molten resin M**1** can

be reduced, and thus a molded article of a good quality can be manufactured. In addition, it is preferable that either one or both of a contact portion of the plunger **55** and a contact portion of the pressing member **711** are spherical surfaces. As a result of this, even if the pressing member **711** moves forward in a slightly inclined state, the plunger **55** can move forward while correcting the inclination of the pressing member **711** so as to follow the inner wall of the cylinder. Therefore, the plunger **55** being caught by the inner wall of the cylinder, that is, occurrence of galling can be reduced.
(85) To be noted, although it is preferable that the plunger **55** and the driving mechanism **71** are separated, the configuration is not limited to this, and the plunger **55** and the driving mechanism **71** may be connected. In addition, the plunger **55** may include a head portion that comes into contact with the molten resin, and a shaft portion separate from the head portion, and the shaft portion may be connected to the driving mechanism **71**.
(86) To be noted, the configuration and operation of members related to the resin channel **522** and the configuration and operation of members related to the resin channel **523** are substantially the same as the configuration and operation of members related to the resin channel **521**. The valve pin **61**, the plunger **55**, and the driving mechanism **71** are members related to the resin channel **521**. The valve pin **62**, the plunger **56**, and the driving mechanism **72** are members related to the resin channel **522**. The valve pin **63**, the plunger **57**, and the driving mechanism **73** are members related to the resin channel **523**. Therefore, molten resin is stably discharged from the discharge ports G**1**, G**2**, and G**3**.
(87) Since the plungers **55**, **56**, and **57** are mutually individually driven by the driving portion **70**, position control thereof can be individually performed, and the capacities of the reserving portions **58**, **59**, and **60** can be individually adjusted. As a result, the injection amount of resin can be adjusted for each of the discharge ports G**1**, G**2**, and G**3**. In addition, the plungers **55**, **56**, and **57** can stably discharge molten resin without being affected by the injection operation of each other.
(88) In addition, the molten resin flowing back to the plasticizing portion **49** can be effectively suppressed at the valve pins **61**, **62**, and **63**. In addition, the opening/closing timings of the valve pins **61**, **62**, and **63** affecting each other can be suppressed, and thus molten resin can be injected by a stable amount.
(89) Here, as illustrated in FIG. **4**, a pressure sensor **64** used for detecting the resin pressure in the partial channel **52***e* is provided in the partial channel **52***e*, and a pressure sensor **65** used for detecting the resin pressure in the partial channel **52***f* is provided in the partial channel **52***f*. The pressure sensor **64** is an example of a first pressure sensor. The pressure sensor **65** is an example of a second pressure sensor. To be noted, an unillustrated pressure sensor used for detecting the resin pressure in the partial channel **52***g* is provided in the partial channel **52***g*. The unillustrated pressure sensor is an example of a third pressure sensor. These pressure sensors are preferably disposed at positions where the sectional area in the channel is small.
(90) The control apparatus **200** controls the driving of the plunger **55** by the driving portion **70** on the basis of a pressure value obtained by detection by the pressure sensor **64**. Similarly, the control apparatus **200** controls the driving of the plunger **56** by the driving portion **70** on the basis of a pressure value obtained by detection by the pressure sensor **65**. Similarly, the control apparatus **200** controls the driving of the plunger **57** by the driving portion **70** on the basis of a pressure value obtained by detection by the unillustrated pressure sensor.
(91) In the first embodiment, the lengths of the branch channels **52***b*, **52***c*, and **52***d* are different from each other. For example, the length L**1** and the length L**2** illustrated in FIG. **4** are different from each other. Therefore, it can be considered that the timings at which the molten resin is respectively injected into the reserving portions **58**, **59**, and **60** also differ. Therefore, according to the first embodiment, by measuring the pressures in the vicinity of the reserving portions **58**, **59**, and **60**, injection conditions corresponding to the pressure values can be set. In addition, by measuring the injection pressure at the time of injection by each of the plungers **55**, **56**, and **57**, the

driving conditions of the plungers **55**, **56**, and **57** can be set such that the injection pressures are equal. In the first embodiment, since the plungers **55**, **56**, and **57** and the driving mechanisms **71**, **72**, and **73** are provided in correspondence with the discharge ports G**1**, G**2**, and G**3**, the driving conditions of the plungers **55**, **56**, and **57** can be independently set. Therefore, the respective injection states of the molten resin injected from the discharge ports G**1**, G**2**, and G**3** can be caused to match each other.
(92) In the first embodiment, the cavities CV**1**, CV**2**, and CV**3** are defined between the workpiece **13** and molds **21**, **22**, and **23** by bringing the molds **21**, **22**, and **23** into contact with the workpiece **13** molded from resin. The mold **21** is an example of a first mold. The mold **22** is an example of a second mold. The mold **23** is an example of a third mold. By injecting molten resin into the cavities CV**1**, CV**2**, and CV**3**, three molded articles are outsert-molded on the workpiece **13**, and thus a product is manufactured.
(93) FIGS. **13** and **14** are explanatory diagrams of a product **10** according to the first embodiment. FIG. **13** is a plan view of the product **10**, and FIG. **14** is a section view of the product **10**. In the first embodiment, members **14**, **15**, and **16** that are resin molded articles are outsert-molded on the workpiece **13**, and thus a product **10** is manufactured. That is, the product **10** is manufactured by adding the members **14**, **15**, and **16** to the workpiece **13**. The member **14** is a molded article molded in the cavity CV**1**, the member **15** is a molded article molded in the cavity CV**2**, and the member **16** is a molded article molded in the cavity CV**3**. The members **14**, **15**, and **16** are, for example, sealing members for sealing powder.
(94) The member **14** has a different volume from the members **15** and **16**. Therefore, the cavity CV**1** of the mold **21** used for molding the member **14** has a different capacity from the cavity CV**2** of the mold **22** used for molding the member **15**. Similarly, the cavity CV**1** of the mold **21** used for molding the member **14** has a different capacity from the cavity CV**3** of the mold **23** used for molding the member **16**.
(95) The resin material used for the molding is preferably a resin that is suitable for a sealing member and can be molded by injection molding, such as a thermoplastic elastomer resin.
(96) In the first embodiment, the positions of the plungers **55**, **56**, and **57** can be independently controlled, and the capacities of the reserving portions **58**, **59**, and **60** can be individually changed. That is, the resin reserving amounts of the reserving portions **58**, **59**, and **60** can be individually adjusted. As a result of this, molten resin can be stably injected from each of the discharge ports G**1**, G**2**, and G**3**, and the quality of the product **10** can be improved.
(97) A manufacturing method for outsert-molding the members **14**, **15**, and **16** on the workpiece **13** will be briefly described. First, the workpiece **13** is attached to the holding member **30** illustrated in FIG. **1**. The molds **21**, **22**, and **23** illustrated in FIG. **8** are disposed on the holding member **30**.
(98) An unillustrated conveyance stage on which the holding member **30** is placed can move in a mold clamping direction and a mold opening direction. The molds are clamped when the conveyance stage is moved in the mold clamping direction. As a result of the mold clamping, the discharge ports G**1**, G**2**, and G**3** are connected to the cavities CV**1**, CV**2**, and CV**3**.
(99) After the mold clamping, in the injection step, the control apparatus **200** moves the valve pins **61**, **62**, and **63** to open the discharge ports G**1**, G**2**, and G**3**. Then, the control apparatus **200** moves the plungers **55**, **56**, and **57** to reduce the capacities of the reserving portions **58**, **59**, and **60**. As a result of this, the molten resin injected into the reserving portions **58**, **59**, and **60** is injected into the cavities CV**1**, CV**2**, and CV**3** from the discharge ports G**1**, G**2**, and G**3**.
(100) In the amount measuring step, the control apparatus **200** moves the valve pins **61**, **62**, and **63** to close the discharge ports G**1**, G**2**, and G**3**. In addition, the control apparatus **200** retracts the pressing members **711**, **712**, and **713** by a predetermined stroke in accordance with the capacities of the cavities CV**1**, CV**2**, and CV**3**. When molten resin is supplied to the reserving portions **58**, **59**, and **60** from the plasticizing portion **49**, the plungers **55**, **56**, and **57** retract by being pressed by the molten resin. When the capacities of the reserving portions **58**, **59**, and **60** reach the capacities of

the cavities CV**1**, CV**2**, and CV**3**, the plungers **55**, **56**, and **57** come into contact with the pressing members **711**, **712**, and **713**. As a result of this, the amount measurement of the molten resin to be injected in the next injection step is completed.

(101) In addition, after the injection step described above, a cooling step of cooling and solidifying the molten resin injected into the cavities CV**1**, CV**2**, and CV**3** is executed in parallel with the amount measuring step. The discharge ports G**1**, G**2**, and G**3** are closed in the amount measuring step, and by opening the molds in this state, the product **10** in which the members **14**, **15**, and **16** are formed on the workpiece **13** is taken out. By repeating the above operation, the products **10** are sequentially and efficiently manufactured. Further, since molten resin can be stably injected into the cavities CV**1**, CV**2**, and CV**3**, the quality of the product **10** to be manufactured is improved.

Second Embodiment

(102) A second embodiment will be described. In the second embodiment, description of elements substantially the same as in the first embodiment will be simplified or omitted. FIG. **15** is a perspective view of a manufacturing system **2000** according to the second embodiment. In the second embodiment, a case where the manufacturing system **2000** includes a plurality of manufacturing apparatuses will be described.

(103) The manufacturing system **2000** includes the manufacturing apparatus **1000** described in the first embodiment, a manufacturing apparatus **1000**A, and a manufacturing apparatus **1000**B. Similarly to the manufacturing apparatus **1000**, the manufacturing apparatus **1000**A includes a plurality of resin channels, and a plurality of plungers corresponding to the plurality of reserving portions. For example, the manufacturing apparatus **1000**A includes two resin channels, and two plungers corresponding to two reserving portions. The two resin channels each has an discharge port. To be noted, the manufacturing apparatus **1000**B includes one resin channel, and one plunger corresponding to one reserving portion. That is, the manufacturing system **2000** includes two manufacturing apparatuses including a plurality of plungers.

(104) The manufacturing apparatus **1000**A includes a resin supply portion **46**A, a plasticizing portion **49**A, a defining portion **552**A, and a driving portion **70**A. The manufacturing apparatus **1000**B includes a resin supply portion **46**B, a plasticizing portion **49**B, a defining portion **552**B, and a driving portion **70**B. The driving portion **70**A is capable of driving the two plungers included in the manufacturing apparatus **1000**A individually or in an interlocked manner. The driving portion **70**B is capable of driving the one plunger included in the manufacturing apparatus **1000**B.

(105) To be noted, although the illustration thereof is omitted, the manufacturing system **2000** includes a control apparatus corresponding to the manufacturing apparatus **1000**, a control apparatus corresponding to the manufacturing apparatus **1000**A, and a control apparatus corresponding to the manufacturing apparatus **1000**B, and these control apparatuses constitute a control system. The control system may be constituted by a single or a plurality of computers.

(106) FIG. **16** is an explanatory diagram of a product **10**A according to the second embodiment. FIG. **16** is a plan view of the product **10**A. The manufacturing apparatus **1000** molds the members **14**, **15**, and **16**, the manufacturing apparatus **1000**A molds a member **17**, and the manufacturing apparatus **1000**B molds a member **18**. The member **17** is a member having an elongated shape, and is formed by injecting molten resin into a cavity connected to a plurality of discharge ports in the manufacturing apparatus **1000**A. The members **17** and **18** can be molded from a different resin material from the members **14**, **15**, and **16**.

(107) As described above, also in the manufacturing apparatus **1000**A, the stroke amount of the plurality of plungers can be appropriately set in accordance with the shape of the member **17** and the flowing characteristics of the resin when molding the member **17**, and thus the molten resin can be stably injected. As a result of this, the quality of the product **10**A to be manufactured is improved.

Third Embodiment

(108) A third embodiment will be described. In the third embodiment, description of elements

substantially the same as in the first embodiment will be simplified or omitted. FIG. **17** is a plan view of part of a manufacturing apparatus according to a third embodiment.

(109) The manufacturing apparatus of the third embodiment includes three plungers **155**C, **156**C, and **157**C defined by the defining portion **52**, branching from the plasticizing portion **49**, and configured to inject resin, and a driving portion **70**C. The driving portion **70**C includes a driving mechanism **171**C that drives the plungers **155**C and **157**C in an interlocked manner, and a driving mechanism **172**C that drives the plunger **156**C independently from the plungers **155**C and **157**C. The plungers **155**C and **157**C are both driven by the driving mechanism **171**C.

(110) If the capacities of two cavities into which molten resin is injected by the two plungers **155**C and **157**C are equal, the two plungers **155**C and **157**C can be driven by the single driving mechanism **171**C.

(111) In the example of the third embodiment, the capacity of the cavity into which molten resin is injected by the plunger **156**C is different from the capacity of each of the two cavities into which the molten resin is injected by the two plungers **155**C and **157**C. In such a case, the plunger **156**C may be configured to be driven by the driving mechanism **172**C different from the driving mechanism **171**C.

(112) As described above, the stroke of each of the plungers **155**C, **156**C, and **157**C can be appropriately set in consideration of the shape of the molded article to be molded and the flowing characteristics of the molten resin. Further, the molten resin injected into a reserving portion corresponding to each of the plungers **155**C and **157**C can be stably injected by a constant stroke.

(113) To be noted, in the case where it is necessary to vary the injection amount between the plungers **155**C and **157**C, the injection amount can be adjusted by disposing a spacer between the plunger **155**C or **157**C and the driving mechanism **171**C. For example, if the amount is measured in a state in which the spacer is disposed between the plunger **155**C and the driving mechanism **171**C and the spacer is removed at the time of injection, the injection amount can be varied between the plungers **155**C and **157**C. Alternatively, for example, if the amount is measured in a state in which the spacer is not disposed between the plunger **157**C and the driving mechanism **171**C, and the injection is performed in a state in which the spacer is disposed between the plunger **157**C and the driving mechanism **171**C at the time of injection, the injection amount can be varied between the plungers **155**C and **157**C.

Fourth Embodiment

(114) A fourth embodiment will be described. In the fourth embodiment, description of elements substantially the same as in the first embodiment will be simplified or omitted. FIG. **18** is a plan view of part of a manufacturing apparatus according to a fourth embodiment.

(115) The manufacturing apparatus of the fourth embodiment includes four plungers **155**D, **156**D, **157**D, and **158**D defined by the defining portion **52**, branching from the plasticizing portion **49**, and configured to inject resin, and a driving portion **70**D. The driving portion **70**D includes a driving mechanism **171**D that drives the plungers **155**D and **157**D in an interlocked manner, and a driving mechanism **172**D that drives the plungers **156**D and **158**D in an interlocked manner. The driving mechanism **171**D drives the plungers **155**D and **157**D independently from the plungers **156**D and **158**D. The driving mechanism **172**D drives the plungers **156**D and **158**D independently from the plungers **155**D and **157**D. The plungers **155**D and **157**D are both driven by the driving mechanism **171**D, and the plungers **156**D and **158**D are both driven by the driving mechanism **172**D.

(116) If the capacities of two cavities into which molten resin is injected by the two plungers **155**D and **157**D are equal, the two plungers **155**D and **157**D can be driven by the single driving mechanism **171**D.

(117) If the capacities of two cavities into which molten resin is injected are equal, the two plungers **156**D and **158**D can be driven by the single driving mechanism **172**D.

(118) In the example of the fourth embodiment, the capacity of each of the two cavities into which

molten resin is injected by the plungers **156**D and **158**D is different from the capacity of each of the two cavities into which the molten resin is injected by the two plungers **155**D and **157**D. In such a case, the plungers **156**D and **158**D may be configured to be driven by the driving mechanism **172**D different from the driving mechanism **171**D.

(119) As described above, the stroke of each of the plungers **155**D, **156**D, **157**D and **158**D can be appropriately set in consideration of the shape of the molded article to be molded and the flowing characteristics of the molten resin. Further, the molten resin injected into a reserving portion corresponding to each of the plungers **155**D and **157**D can be stably injected by a constant stroke. In addition, the molten resin injected into a reserving portion corresponding to each of the plungers **156**D and **158**D can be stably injected by a constant stroke.

(120) To be noted, in the case where it is necessary to vary the injection amount between the plungers **155**D and **157**D, the injection amount can be adjusted by disposing a spacer between the plunger **155**D or **157**D and the driving mechanism **171**D.

(121) To be noted, in the case where it is necessary to vary the injection amount between the plungers **156**D and **158**D, the injection amount can be adjusted by disposing a spacer between the plunger **156**D or **158**D and the driving mechanism **172**D.

Fifth Embodiment

(122) A fifth embodiment will be described. In the fifth embodiment, description of elements substantially the same as in the first embodiment will be simplified or omitted. FIG. **19** is a plan view of part of a manufacturing apparatus according to a fifth embodiment.

(123) The manufacturing apparatus of the fifth embodiment includes three plungers **155**E, **156**E, and **157**E defined by a defining portion, branching from a plasticizing portion, and configured to inject resin, and a driving portion **70**E. The driving portion **70**E is a driving mechanism that drives the plungers **155**E, **156**E, and **157**E in an interlocked manner. The plungers **155**E, **156**E, and **157**E are all driven by the driving portion **70**E.

(124) If the capacities of three cavities into which molten resin is injected by the three plungers **155**E to **157**E are equal, the three plungers **155**E to **157**E can be driven by the single driving portion **70**E.

(125) As described above, the stroke of each of the plungers **155**E to **157**E can be appropriately set in consideration of the shape of the molded article to be molded and the flowing characteristics of the molten resin. Further, the molten resin injected into a reserving portion corresponding to each of the plungers **155**E to **157**E can be stably injected by a constant stroke.

(126) To be noted, in the case where it is necessary to vary the injection amount between the plungers **155**E to **157**E, the injection amount can be adjusted by disposing a spacer between any of the plungers and the driving portion **70**E.

Sixth Embodiment

(127) A sixth embodiment will be described. In the sixth embodiment, description of elements substantially the same as in the first embodiment will be simplified or omitted. FIG. **20** is an explanatory diagram of two resin channels **521**F and **522**F of a manufacturing apparatus according to the sixth embodiment. The configuration of the resin channels in the sixth embodiment is different from the first embodiment.

(128) The manufacturing apparatus of the sixth embodiment includes the plasticizing portion **49**, a defining portion that defines two resin channels **521**F and **522**F, and two plungers **55**F and **56**F. The resin channels **521**F and **522**F do not have a main channel shared therebetween, and is configured to directly branch from the supply port S**1** connected to the plasticizing portion **49**.

(129) The resin channel **521**F is an example of a first resin channel connecting the supply port S**1** connected to the plasticizing portion **49** and the discharge port G**1**. The resin channel **522**F is an example of a second resin channel connecting the supply port S**1** connected to the plasticizing portion **49** and the discharge port G**2**. Similarly to the first embodiment, the discharge port G**1** is connected to the cavity CV**1**, and the discharge port G**2** is connected to the cavity CV**2**. The

plunger **55**F is an example of a first plunger. The plunger **56**F is an example of a second plunger.
(130) The resin channel **521**F includes a reserving portion **58**F, a partial channel **531**F, and a partial channel **532**F. The reserving portion **58**F is an example of a first reserving portion, and is a space capable of reserving molten resin supplied from the supply port S**1** connected to the plasticizing portion **49**. The partial channel **531**F is an example of a first partial channel, and is a channel connecting the reserving portion **58**F and the supply port S**1** connected to the plasticizing portion **49**. The partial channel **532**F is an example of a second partial channel, and is a channel connecting the reserving portion **58**F and the discharge port G**1**.
(131) The resin channel **522**F includes a reserving portion **59**F, a partial channel **533**F, and a partial channel **534**F. The reserving portion **59**F is an example of a second reserving portion, and is a space capable of reserving molten resin supplied from the supply port S**1** connected to the plasticizing portion **49**. The partial channel **533**F is an example of a third partial channel, and is a channel connecting the reserving portion **59**F and the supply port S**1** connected to the plasticizing portion **49**. The partial channel **534**F is an example of a fourth partial channel, and is a channel connecting the reserving portion **59**F and the discharge port G**2**.
(132) The plunger **55**F is capable of moving to reduce the capacity of the reserving portion **58**E The plunger **55**F moves to reduce the capacity of the reserving portion **58**F, and thus the molten resin reserved in the reserving portion **58** is injected into the cavity CV**1** from the discharge port G**1** via the partial channel **532**F.
(133) The plunger **56**F is capable of moving to reduce the capacity of the reserving portion **59**F. The plunger **56**F moves to reduce the capacity of the reserving portion **59**F, and thus the molten resin reserved in the reserving portion **59** is injected into the cavity CV**2** from the discharge port G**2** via the partial channel **534**F.
(134) Also in the configuration of the resin channels **521**F and **522**F described above, similarly to the first embodiment, the molten resin can be stably injected from each of the discharge ports G**1** and G**2**, and thus the quality of the product to be manufactured can be improved.
(135) As described above, according to the present disclosure, the quality of the product to be manufactured is improved.
(136) The present disclosure is not limited to the embodiments described above, and embodiments can be modified in many ways within the technical concept of the present disclosure. In addition, the effects described in the embodiments are merely enumeration of the most preferable effects that can be obtained from embodiments of the present disclosure, and effects of embodiments of the present disclosure are not limited to those described in the embodiments.
(137) For example, the manufacturing apparatus may be configured such that a plurality of discharge ports are respectively connected to a plurality of cavities, or a plurality of discharge ports are connected to one cavity. The plurality of cavities may be defined by one mold, or may be respectively defined by a plurality of molds. In addition, part of each of the plurality of cavities or part of the single cavity may be defined by a workpiece.
(138) Furthermore, the contents of disclosure in the present specification include not only contents described in the present specification but also all of the items which are understandable from the present specification and the drawings accompanying the present specification. Moreover, the contents of disclosure in the present specification include a complementary set of concepts described in the present specification. Thus, if, in the present specification, there is a description indicating that, for example, “A is B”, even when a description indicating that “A is not B” is omitted, the present specification can be said to disclose a description indicating that “A is not B”. This is because, in a case where there is a description indicating that “A is B”, taking into consideration a case where “A is not B” is a premise.
(139) While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all

such modifications and equivalent structures and functions.
(140) This application claims the benefit of Japanese Patent Application No. 2022-46727, filed Mar. 23, 2022, which is hereby incorporated by reference herein in its entirety.

Claims

1. A manufacturing apparatus comprising: a defining portion configured to define (i) a supply port through which molten resin is supplied, (ii) a discharge port through which molten resin is discharged, and (iii) a resin channel connecting the supply port and the discharge port, the resin channel including a reserving portion configured to reserve molten resin, a first partial channel through which molten resin from the supply port passes, a second partial channel connected to the discharge port and through which molten resin from the reserving portion passes, a relay channel connected to the second partial channel and the reserving portion and through which molten resin that has passed through the first partial channel passes, and a connecting channel connecting the first partial channel and the second partial channel; a plunger that is movable to change a capacity of the reserving portion and configured to discharge molten resin reserved in the reserving portion from the discharge port by moving to reduce the capacity of the reserving portion; and a valve member provided in the connecting channel and the second partial channel and movable in a channel direction of the second partial channel, wherein the valve member includes: a first valve portion configured to open and close the discharge port by moving in the channel direction; and a second valve portion configured to, in a case where the first valve portion has moved to a position to close the discharge port, open a resin exit of the first partial channel such that the first partial channel communicates with the relay channel allowing molten resin to move into the reserving portion, and in a case where the first valve portion has moved to a position to open the discharge port, close the resin exit of the first partial channel allowing molten resin in the reserving portion to move to the discharge port.
2. The manufacturing apparatus according to claim 1, wherein the second valve portion has an outer peripheral surface that is in contact with an inner peripheral surface of the defining portion so as to be slidable in the channel direction, and that is configured to open and close the resin exit of the first partial channel by moving in the channel direction.
3. A manufacturing apparatus comprising: a defining portion configured to define (i) a supply port through which molten resin is supplied, (ii) a first discharge port through which molten resin is discharged, (iii) a second discharge port through which molten resin is discharged, (iv) a first resin channel connecting the supply port and the first discharge port, and (v) a second resin channel connecting the supply port and the second discharge port, the first resin channel including a first reserving portion configured to reserve molten resin, the second resin channel including a second reserving portion configured to reserve molten resin; a first plunger that is movable to change a capacity of the first reserving portion and configured to discharge molten resin reserved in the first reserving portion from the first discharge port by moving to reduce the capacity of the first reserving portion; a second plunger that is movable to change a capacity of the second reserving portion and configured to discharge molten resin reserved in the second reserving portion from the second discharge port by moving to reduce the capacity of the second reserving portion; a first valve member; and a second valve member, wherein the first resin channel includes a first partial channel through which molten resin from the supply port passes, a second partial channel connected to the first discharge port and through which molten resin from the first reserving portion passes, and a first relay channel connected to the second partial channel and the first reserving portion and through which molten resin that has passed through the first partial channel passes, the second resin channel includes a third partial channel through which molten resin from the supply port passes, a fourth partial channel connected to the second discharge port and through which molten resin from the second reserving portion passes, and a second relay channel connected

to the fourth partial channel and the second reserving portion and through which molten resin that has passed through the third partial channel passes, the first plunger discharges molten resin reserved in the first reserving portion from the first discharge port via the first relay channel and the second partial channel by moving to reduce a capacity of the first reserving portion, the second plunger discharges molten resin reserved in the second reserving portion from the second discharge port via the second relay channel and the fourth partial channel by moving to reduce a capacity of the second reserving portion, the first valve member includes a first valve portion configured to open and close the first discharge port in a case where the first valve member moves, and a second valve portion configured to, in a case where the first valve portion has moved to a position to close the first discharge port, open a resin exit of the first partial channel such that the first partial channel communicates with the first relay channel allowing molten resin to move into the first reserving portion, and in a case where the first valve portion has moved to a position to open the first discharge port, close the resin exit of the first partial channel allowing molten resin in the first reserving portion to move to the first discharge port, the second valve member includes a third valve portion configured to open and close the second discharge port in a case where the second valve member moves, and a fourth valve portion configured to, in a case where the third valve portion has moved to a position to close the second discharge port, open a resin exit of the third partial channel such that the third partial channel communicates with the second relay channel allowing molten resin to move into the second reserving portion, and in a case where the third valve portion has moved to a position to open the second discharge port, close the exit of the third partial channel allowing molten resin in the second reserving portion to move to the second discharge port, and the first partial channel is longer than the third partial channel.
4. The manufacturing apparatus according to claim 3, wherein the first relay channel is defined such that a sectional area thereof increases toward the first reserving portion, and the second relay channel is defined such that a sectional area thereof increases toward the second reserving portion.
5. The manufacturing apparatus according to claim 3, wherein the first resin channel includes a connecting channel connecting the first partial channel and the second partial channel, the first valve member is provided in the connecting channel and the second partial channel and is movable in a channel direction of the second partial channel, the first partial channel and the first relay channel are connected to the connecting channel so as to intersect with the connecting channel, and the first valve member opens and closes the first discharge port by moving in the channel direction.
6. The manufacturing apparatus according to claim 5, wherein the second valve portion includes an outer peripheral surface that is in contact with an inner peripheral surface of the defining portion so as to be slidable in the channel direction and that opens and closes the resin exit of the first partial channel by moving in the channel direction.
7. The manufacturing apparatus according to claim 3, further comprising: a driving portion configured to individually drive the first plunger and the second plunger; a first pressure sensor configured to detect a resin pressure in the first relay channel; a second pressure sensor configured to detect a resin pressure in the second relay channel; and a controller configured to control driving of the first plunger by the driving portion on a basis of a pressure value obtained by detection by the first pressure sensor, and configured to control driving of the second plunger by the driving portion on a basis of a pressure value obtained by detection by the second pressure sensor.
8. The manufacturing apparatus according to claim 3, wherein a total length of the first partial channel and the first relay channel is different from a total length of the third partial channel and the second relay channel.
9. The manufacturing apparatus according to claim 3, further comprising: a first heater disposed along the second partial channel; and a second heater disposed along the fourth partial channel, wherein the first heater is longer than the second heater.
10. The manufacturing apparatus according to claim 3, wherein the first partial channel and the third partial channel are each connected to a main channel shared by the first resin channel and the

second resin channel and connected to the supply port, the first partial channel includes a first branch channel branching from the main channel so as to be not linearly connected to the main channel, and the third partial channel includes a second branch channel branching from the main channel so as to be not linearly connected to the main channel.

11. The manufacturing apparatus according to claim 3, further comprising a driving portion configured to individually drive the first plunger and the second plunger.

12. The manufacturing apparatus according to claim 3, further comprising a driving portion configured to drive the first plunger and the second plunger in an interlocked manner.

13. The manufacturing apparatus according to claim 3, wherein the defining portion further defines a third resin channel including a third reserving portion configured to reserve molten resin and connecting the supply port and a third discharge port, and the manufacturing apparatus further comprises: a third plunger that is movable to change a capacity of the third reserving portion and configured to discharge molten resin reserved in the third reserving portion from the third discharge port by moving to reduce the capacity of the third reserving portion; and a driving portion configured to drive the first plunger and the third plunger in an interlocked manner.

14. The manufacturing apparatus according to claim 3, further comprising: a first pressing member capable of being separated from the first plunger and capable of pressing the first plunger so as to reduce a capacity of the first reserving portion; and a second pressing member capable of being separated from the second plunger and capable of pressing the second plunger so as to reduce a capacity of the second reserving portion.

15. The manufacturing apparatus according to claim 3, further comprising a plasticizing portion connected to the supply port and configured to melt resin.

16. A manufacturing system comprising: the manufacturing apparatus according to claim 15; and a plurality of plasticizing portions including the plasticizing portion.

17. The manufacturing apparatus according to claim 1, wherein the first partial channel and the relay channel are connected to the connecting channel so as to intersect with the connecting channel.

18. The manufacturing apparatus according to claim 1, wherein the valve member is moved back and forth in one direction of the channel direction to control whether to allow molten resin to move to the reserving portion or the discharge port.

19. The manufacturing apparatus according to claim 1, further comprising a plasticizing portion connected to the supply port and configured to melt resin, wherein the first valve portion moves to a position that opens the discharge port to close the resin exit of the first partial channel, thereby suppressing molten resin from flowing back into the plasticizing section.

20. The manufacturing apparatus according to claim 1, wherein the valve member defines a space for the first partial channel to communicate with the relay channel in the case where the first valve portion is moved to the position to close the discharge port.

21. A manufacturing method for a product, the manufacturing method comprising manufacturing the product by the manufacturing apparatus according to claim 1.

22. A manufacturing method for a product, the manufacturing method comprising manufacturing the product by the manufacturing apparatus according to claim 3, wherein resin discharged from the first discharge port and resin discharged from the second discharge port are added to a single workpiece.

23. A manufacturing method for a product, the manufacturing method comprising manufacturing the product by the manufacturing system according to claim 16.